

Statement of Work (SOW) for GRCop-42 Hardware

A requirement of the HALO Engines Space Act Agreement is to partner with an additive manufacturing (AM) supplier that is able to explicitly meet the production requirements for powder bed fusion components. The National Aeronautics and Space Administration (NASA), specifically Marshall Space Flight Center (MSFC) has a need for fabrication and post-processing of additively manufactured rotating detonation rocket engine hardware. Specifically, combustion chambers and injectors designed for the 5K lbf class. This hardware must be designed and fabricated with no witness lines, restarts, or coater arm tears. Moreover, all passages must be free of powder. All parts must be post processed in a specific order and with specific processing criteria. Thus, a sole source selection of this work is being submitted for procurement.

Task 1, Production of Additively Manufactured Hardware

Vendor shall produce a single outer body and single CORE component. All hardware will be designed by NASA engineers at Marshall Space Flight Center and their respective models supplied to vendor before production. During production the vendor must meet the following requirements. Provide all hardware with no witness lines, restarts, or coater arm tears. If vendor is unable to provide a component with no powder remaining internal to the part, a reprint will be required at no additional cost.

Task 2, Post Processing of Additively Manufactured Components

Several of the required post processing steps shall be conducted by the vendor. All hardware shall be de-powdered, and no passages shall be blocked with powder as shown by CT scanning. Parts will be sent to a mutually agreed vendor for Hot Isostatic Pressing (HIP) operations. Vendor will remove the part from the build plate using electrical discharge machining (EDM). Vacuum nucleation will also be conducted to clean the parts. Finally, chemical mechanical polishing shall be conducted at a select vendor and coordinated by the hardware vendor.

Requirements for Additively Manufactured GRCop-alloy hardware:

1. Vendor shall fabricate all chamber and injector hardware using powder bed fusion (PBF) additive manufacturing.
2. Vendor shall provide material to complete all builds.
3. Density of the material in the as-built condition shall be a minimum of 99.2% prior to any post-processing heat treatments or Hot Isostatic Pressing (HIP) operations and be demonstrated prior to fabrication of any hardware.
4. Vendor shall complete build in one-piece and remove all powder from the channels prior to further operations defined in Requirement #6.
5. Part shall not be exposed to moisture until powder from channels can be confirmed.
6. Part shall complete Hot Isostatic Pressing (HIP) according to NASA processing parameters. Part shall remain on the build plate for HIP operations to help with handling.
7. As-built surface finish requested, no media blasting.
8. Any anomalies during the build process of the final part should be reported to MSFC Technical Point of Contact.
9. Vendor shall remove parts from the build plate prior to delivery.
10. Parts shall be delivered to Marshall Space Flight Center.
11. Vendor shall provide a basic build package including any build anomalies and data relative as evidenced of process validation under documentation required by AS9100, ISO9001, or similar standard.
12. Vendor shall be ITAR registered or demonstrate ability to handle ITAR data.

Shipping

1. Contractor shall ship any applicable hardware deliverables to:

Central Receiving, Bldg 4631

[REDACTED]

4203:4312

NASA Marshall Space Flight Center

Huntsville, AL 35812

Acronyms:

CT = Computer Tomography

EDM = Electrodischarge Machining

GRCop = Glenn Research Center Copper-Alloy (Cu-Cr-Nb)

HIP = Hot Isostatic Pressing

ITAR = International Traffic in Arms Regulations

MSFC = Marshall Space Flight Center

NLT = No Later Than

PBF = Powder Bed Fusion